Mathematical Monism from an Exhaustive Dichotomy

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Abstract

Anything that exists is either patterned or random. Both are mathematical: patterns are computable functions; randomness is defined by Kolmogorov complexity. The pattern-randomness dichotomy (PRD) is exhaustive. No coherent third category exists. Any proposed non-mathematical substance must have determinate features to differ from nothing, and those features are patterned or random. The escape routes all lead back.

From this foundation: all consistent mathematical structures exist, because any filter selecting among them is itself a consistent structure, and all filters exist, including the null filter. Among all structures, simpler ones dominate the probability measure exponentially (the Simplicity Dominance Principle). Five consequences follow: Occam's Razor becomes a theorem; the arrow of time is grounded in computational irreversibility; cosmological fine-tuning dissolves; the hard problem of consciousness assumes its conclusion; and three pillars of physics, the Principle of Least Action, the arrow of time, and emergence, reduce to one structural fact. The position is falsifiable: if the final theory of physics requires irreducibly high complexity, or if physical laws do not take variational form, it is challenged.

Keywords: mathematical monism, pattern-randomness dichotomy, Kolmogorov complexity, structural realism, Simplicity Dominance Principle

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1. The Proof

1.1 Existence requires determinacy

For anything to exist and be distinguishable from nothing, it must possess determinate features. Something differs from nothing by having properties: spatial extension, charge, mass, complexity, duration. If these are entirely undefined, the something is indistinguishable from nothing. This is not a claim about epistemology, about what we can know. It is a claim about ontology: what it is for something to exist. Existence requires determinacy. Determinacy requires distinguishing features. Distinguishing features require relations of sameness and difference.

Properties do not float free. Each is defined by its connections to others. Mass relates to energy through $E=mc^2$. Position relates to velocity through derivative relationships. A property's identity is constituted by how it relates to other properties within the overall structure. Properties form a relational structure.

Physics has progressively confirmed this. What seemed like solid particles became excitations in fields. Fields became representations of symmetry groups. Forces became geometric curvatures in spacetime. At each level, the concrete substance dissolved into relational structure. The electron is not a tiny ball with properties attached. It is a pattern of relationships: how it couples to the electromagnetic field, how it responds to measurements, how it relates to other fermions through the Pauli exclusion principle. Remove these relations and nothing remains. Structure is the substance. The Standard Model is purely relational: gauge fields, coupling constants, symmetry groups. No underlying stuff.

1.2 The dichotomy

Determinate features are either patterned or random. This is made precise by algorithmic information theory. A pattern is a structure whose Kolmogorov complexity K(S) is strictly less than its length: there exists a rule simpler than the data that generates it. The laws of physics, the digits of pi, fractal coastlines, deterministic chaos are all patterns. Short programs generating vast complexity.

Randomness is the complement: K(S) is approximately equal to the length of S. The sequence is algorithmically incompressible. No shorter description exists. Crucially, randomness is not the absence of mathematics. Kolmogorov complexity is itself a mathematical function. The statement "this sequence is random" is a mathematical claim about a mathematical object. Even structurelessness is structure, since it is defined and measured mathematically.

Since K is a well-defined function mapping any finite string to an integer, both categories are strictly mathematical objects. Every finite sequence is either compressible or incompressible. There is no third option. The dichotomy is exhaustive. This is not a tautology achieved by defining mathematics broadly. The claim that randomness is mathematical is a synthetic discovery of twentieth-century algorithmic information theory, not a trivial restatement.

Any proposed non-mathematical substance must have determinate features, otherwise it is nothing. Those features are patterned or random. Both are mathematical. The proposed substance is mathematical. The escape routes all lead back. Any objection must itself be structured, which is to say: any objection uses structure to deny structure.

1.3 Interaction collapses dualism

Any two things that interact must share structure for causal commerce. If mind and body interact, they share relational structure. Shared structure is mathematical structure. Any interacting dualism, substance, property, or theological, collapses into mathematical monism. A non-structural soul that interacts with a structural body is incoherent: for the interaction itself to have determinate character, both relata must be structural.

Descartes located the interaction at the pineal gland. The problem was not anatomy. The problem was ontology. Two categorically different substances cannot exchange information unless they share a common relational framework. That framework is mathematical structure. Leibniz saw this and proposed pre-established harmony to avoid the interaction problem. Spinoza saw it and proposed one substance. Mathematical monism agrees with Spinoza and derives the conclusion: not Deus sive Natura but Deus sive Mathematica.

The argument applies recursively. Suppose someone posits a non-mathematical property, call it Q, that interacts with mathematical structure. For Q to interact with structure, Q must have features that structure can couple to. Those features are determinate (else the coupling is undefined). Determinate features are patterned or random. Both are mathematical. Q is mathematical. The dualism was apparent. Every proposed non-mathematical layer reduces to mathematics on contact with anything that is mathematical, which is everything.

1.4 The identity argument

Physical reality and its isomorphic mathematical structure share all relational properties. What distinguishes them? Not their relationships: identical by definition of isomorphism. Not their causal powers: causal powers supervene on relational structure. Not their observable properties: all observable properties reduce to relational structure. Color is wavelength. Hardness is bonding strength. Temperature is molecular motion.

The only remaining candidate is non-relational substrate: some primitive stuff that bears properties without being constituted by them. But physics has eliminated it. Quantum field theory contains no objects with intrinsic properties independent of their role in relational structure. And substrate does no explanatory work. Two systems with identical relational structure but hypothetically different substrates behave identically in every observable way. The substrate is explanatorily idle.

Any proposed substrate must itself have determinate features, else it is nothing. Those features are patterned or random. Both are mathematical. Substrate is structure. The regress terminates. By the Identity of Indiscernibles: if two things share all properties, they are identical. Physical reality and its isomorphic mathematical structure share all relational properties. They are not merely similar. They are identical. The map is the territory.

Consider chess implemented in carved wood, in computer memory, or as pure mathematical abstraction. No chess property differs across implementations. No legal move, no winning strategy, no checkmate changes. The game is the relational structure. The wood is merely one access-mode. Burning the wood destroys a token, not the type. The distinction marks a difference in our access, not in the object accessed.

1.5 Transcendental constraint

Observers are logically constrained to patterned realities. Observation requires differentiation, differentiation requires pattern. Memory requires stable encoding, which presupposes regularity. Perception requires distinguishing signal from noise, which presupposes compressibility. An observer in brute randomness is incoherent: a four-sided triangle. This is not probabilistic selection. It is a logical constraint. The question "why do we observe patterns?" is self-answering, since observation presupposes pattern. No world without pattern can contain a mind that notices its own absence.

1.6 Mathematical structure is necessary

Mathematical structure is uncreated and necessary. No agent makes 2+2=4, and none could make it otherwise. No conceivable process could make 2+2 cease to equal 4. The rules are the things. The question "why is there something rather than nothing?" dissolves at necessity rather than brute fact. Mathematical structure cannot fail to exist. Asking what caused it commits a category error. Causation is a relation within structure. Math is what causation is.

Only consistent mathematics forms structure. Inconsistent statements can be written but not constructed. "The set of all sets that do not contain themselves" is syntactically valid and structurally void. You can label a contradiction. You cannot build one. Consistency is the filter on what exists.

2. All Consistent Structures Exist

If all existence is mathematical structure, which structures exist? Any filter selecting among consistent structures is itself a consistent structure. So all filters exist. Including the null filter that excludes nothing. All consistent mathematical structures exist. This is not hypothesis. It is the unique non-arbitrary conclusion.

Tegmark (2008) posits this ensemble as a cosmological conjecture. The PRD derives it. The difference matters. Positing an ensemble invites the objection "why this ensemble rather than another?" Deriving it via the null filter closes that objection. The ensemble is not one hypothesis among alternatives. It is what remains when you ask what could prevent a consistent structure from existing, and find that any answer is itself a structure in the ensemble.

The Level IV multiverse follows as deductive consequence. Every consistent set of laws, every consistent set of initial conditions, every consistent mathematics is realized. Not because the universe is generous. Because nothing could stop it.

3. The Measure

Among all structures, which should observers expect to inhabit? Algorithmic probability (Solomonoff, 1964) provides the natural measure: P(S) is proportional to 2 raised to the negative K(S), where K is Kolmogorov complexity. Each additional bit of specification halves a structure's measure. Simpler structures dominate exponentially. I call this the Simplicity Dominance Principle (SDP).

Solomonoff invented the measure for inductive inference. Schmidhuber (2000) applied it to the universe ensemble. What is new is the derivation: the PRD establishes why the ensemble exists. The null filter shows it is non-arbitrary. Algorithmic probability is its natural measure. Each piece was known. The combination derives what was previously posited.

SDP does not predict we observe the simplest possible physics. The simplest possible structure has no observers. SDP predicts that structures producible by shorter programs have exponentially higher measure, because a short program occupies a larger fraction of program-space. We observe roughly 200 bits of physics, not 10 and not 200,000, because a generating rule of that complexity is short enough to have high measure while complex enough to produce observers. The prediction is statistical: we are more likely to find ourselves in a structure generated by a shorter program, because shorter programs occupy more of program-space.

4. Five Consequences

4.1 Occam's Razor is a theorem

Occam's Razor has been a methodological heuristic for seven centuries. Under SDP it is a theorem. Lower-K structures have exponentially higher measure. Simpler theories describe more probable structures. The success of parsimony in science is not convention or luck. It tracks the measure over all that exists.

Solomonoff's induction already proved this for prediction. SDP extends it to ontology: simpler structures do not merely predict better. They are more real, in the precise sense of occupying more of measure-space. The razor cuts because simpler structures are exponentially more probable. Each bit of unnecessary complexity halves the odds. This is why physics keeps finding that shorter equations describe more of reality. The universe is not obliging physicists. Physics tracks the measure.

4.2 The arrow of time

The standard account grounds the arrow of time in thermodynamics: entropy increases from a low-entropy past. This is the Past Hypothesis, a brute postulate about boundary conditions. SDP provides the derivation: blocks specified by simple initial conditions have lower K and dominate the measure. Observers find themselves in blocks with a simple end and a complex end. The simple end is what they call “past.”

An independent arrow reinforces this. Computation is deterministic forward: 1+1=2. It is underdetermined backward: 2=x+y has infinitely many solutions. Consciousness requires modeling. Modeling requires computation. Therefore consciousness has an inherent direction: the deterministic one. This arrow is independent of entropy, thermodynamics, and boundary conditions. It is a consequence of the logical structure of computation itself. In a block universe where all states exist timelessly, backward states are present. No mind can navigate them as a mind, because no mind can compute in the underdetermined direction. Without consciousness, time is just another math dimension in the block universe. Time is our felt conscious modeling of forward computation. The two arrows, measure-theoretic and computational, converge from independent premises on the same temporal direction.

4.3 Fine-tuning dissolves

The fine-tuning of physical constants is often presented as evidence for a designer or an unexplained coincidence. Under SDP, it dissolves. Observers exist only in structures complex enough to support them. Among those, simpler structures dominate. The constants we observe are the simplest ones compatible with observers. No tuner is needed. No coincidence is required.

The anthropic observation is a selection effect within a derived ensemble, not evidence for anything beyond mathematical necessity. The surprise evaporates once the ensemble is derived rather than posited. In a posited ensemble, fine-tuning asks: “why did someone set the dials?” In a derived ensemble, the dials were never set. All settings exist. Observers find themselves at the settings that produce observers. The simplest such settings dominate.

4.4 The hard problem of consciousness

The hard problem (Chalmers, 1996) asks why physical processes produce subjective experience. The demand is for a non-structural explanation. But any explanation is itself a manipulation of structured symbols. Any concept of consciousness is a structure. Any interaction with consciousness is an interaction with structure. The demand for a non-structural explanation is incoherent: it assumes what it aims to establish, that there is something beyond structure requiring explanation. The hard problem assumes its conclusion.

Consciousness is felt transitions in self-modeling computation. A mind that models itself generates internal states that refer to internal states. This self-reference is not mysterious. It is a property of certain computations, like a program that reads its own source code. The felt quality is what transitions in self-modeling computation are from the inside. A dead brain contains a self-model but does not run it. The threshold question, where does modeling begin, is the central open problem. The philosophical zombies of Chalmers can be labeled but not constructed: labeling a system "structurally identical to a conscious being but lacking experience" presupposes that experience is non-structural. The PRD rules this out. If the structure is identical, the experience is identical. There is nowhere for the difference to hide.

4.5 Many-to-one as structural root

The Principle of Least Action, the arrow of time, and the emergence of smooth physics from discrete rules look like three separate facts. They are one fact: many-to-one mappings.

Computation is many-to-one: 1+1=2 forward, 2=x+y backward. Path integrals are many-to-one: non-extremal paths cancel, the extremal survives. Coarse-graining is many-to-one: many microstates map to one macrostate. All three produce the same result: variational survivors at the scale observers inhabit.

The Principle of Least Action says systems follow extremal paths because non-extremal paths cancel under phase interference. The arrow of time says computation runs one way because many-to-one functions have a deterministic forward direction. Emergence says smooth continuum physics arises from discrete rules because coarse-graining averages over microscopic detail. One structural fact, three consequences. This is not analogy. The path integral literally is a many-to-one map from trajectories to amplitudes. Entropy increase literally is a many-to-one map from microstates to macrostates. The universe minimizes because mathematics does.

5. Falsification

The position is falsifiable through its consequences. If the final theory of physics requires irreducibly high complexity, SDP is challenged. If physical laws do not take variational form, the many-to-one derivation fails. If a force or phenomenon is discovered with no geometric or structural origin, mathematical monism faces an anomaly. If consciousness is demonstrated to depend on non-structural properties, something beyond substrate-independence, the hard-problem dissolution fails.

These are honest constraints, not escape hatches. Call the conclusion tautological: so is 2+2=4. A tautology on reality itself is the deepest possible truth. It is also the most exposed: any counterexample to the dichotomy refutes the entire edifice.

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